MONISH KAMWAL

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EDUCATION

Bachelor of Science **Jan 2023 - Present**

Memorial University of Newfoundland

- Major in Computer Science (co-op) & Minor in Mathematics
- GPA: 3.75 (86.4%)

WORK EXPERIENCE

AI Developer Co-op | Shared Services Canda **Jan 2026 - Dec 2026**

Child Disability Services Chatbot | Manitoba Families Department

- Engineered an enterprise RAG ingestion pipeline using Azure Functions, Blob Storage, and Azure AI Document Intelligence (prebuilt-layout) to crawl, extract, and convert multi-format public assets and complex PDFs into structure-aware Markdown.
- Optimized vector retrieval performance and context relevance by implementing custom header- and table-aware chunking strategies integrated with Azure OpenAI Embeddings and Azure AI Search Vector Indexing.
- Productionized public-facing LLM inference workflows using GPT-4o/mini models; integrated Azure AI Foundry Prompt Guardrails to mitigate jailbreaks, ensure input/output safety, and deliver citation-backed responses.

Fleet Manager Platform V2 | Government of Newfoundland & Labrador

- Orchestrated end-to-end MLOps for Fleet Manager v2, establishing automated data ingestion, Azure DevOps CI/CD pipelines, self-hosted build agents, and an MLflow tracking server for experiment monitoring and model lifecycle management.
- Provisioned and co-managed Azure cloud infrastructure, configuring real-time telemetry, log analytics, and cost-optimization monitoring across all deployed AI microservices.

Recognition & Impact

- Recognized with Technical Innovator Award (Winter 2026) and Problem Solver Award (Spring 2026) for delivering end-to-end government AI solutions.
- Featured on the SSC Gazette & the official Government of Canada Website for driving AI innovation and public sector modernization.

AI Developer Co-op | Public Services & Procurement Canada **Sep 2025 - Dec 2025**

Fleet Manager Platform V1 | Government of Newfoundland & Labrador

- Containerized and deployed the initial Fleet Manager MVP on Azure Virtual Machines using Docker and Caddy reverse proxy to enforce automated HTTPS/TLS, route isolation, and domain management.
- Trained and evaluated baseline ML predictive models using Scikit-learn to forecast fuel consumption rates and vehicle availability trends for fleet operations.
- Authored and standardized the C.L.E.A.N. data preprocessing framework using Pandas and NumPy, establishing team-wide standards for cleaning and structuring raw fleet data.

AGILE Leadership & Recognition

- Facilitated Agile workflows and technical onboarding, serving as Scrum Master for daily standups, designing Figma wireframes, and leading team workshops on Docker and data preprocessing
- Featured on Memorial University's Gazette for public sector software contributions; presented technical solution architectures in executive leadership meetings.

ML Researcher | Visual and Analytic Computing (VAC) Lab **Oct 2024 - Dec 2024**

Breast Cancer Detection and Medical Report Generation

- Co-developed a Vision-Language Model (VLM) with faculty and graduate researchers for automated breast cancer detection and clinical report generation under strict HIPAA/medical compliance frameworks.
- Benchmarked and evaluated 23+ LLM/CVM architectures across 8 datasets, analyzing trade-offs between parameter scale, inference latency, and diagnostic accuracy to inform production selection.
- Fine-tuned and deployed open-source multimodal models, applying context window optimization and model compression techniques for deployment in constrained environments.
- Authored comprehensive technical documentation, delivering the final research report and CAIR manual detailing experimental methodology, model evaluation frameworks, and deployment guidelines.

PROJECTS

QuickDraw MLOps Platform **June 2026 - Nov 2026**

Personal Project • Live Demo & Artifacts

- Engineered a full-stack, automated MLOps platform (PyTorch, ONNX Runtime, DVC, MLflow, FastAPI) running end-to-end retraining, drift evaluation, and serving pipelines under $5/month total cloud overhead.
- Orchestrated automated CI/CD model deployment workflows via GitHub Actions with keyless AWS OIDC authentication, implementing a champion/challenger quality gate (0.85 test accuracy floor) to block degraded models from reaching production.
- Architected multi-tier IaC infrastructure using dual Terraform roots, isolating persistent serverless Lambda API endpoints from ephemeral EKS load-testing clusters managed with Helm, Prometheus, and a 3-layer automated teardown fail-safe.
- Built a closed-loop active learning flywheel and monitoring pipeline, collecting user feedback into an append-only S3 store, retraining models on validated failure cases, and scoring distribution drift using Evidently.

SKILLS

- **AI / ML & Data Science:** PyTorch, ONNX Runtime, Scikit-Learn, Vision-Language Models (VLMs), Transformers, Statistical Modeling (Statsmodels), Pandas, NumPy, Matplotlib
- **MLOps & GenAI Engineering:** RAG Pipelines, Azure AI Search (Vector Indexing), Azure AI Document Intelligence, Azure OpenAI (GPT-4o/mini), Azure AI Foundry (Prompt Guardrails), MLflow, DVC, Pandera, Evidently AI, Active Learning Flywheels
- **Cloud & MLOps Infrastructure:** Azure (Functions, Blob Storage, AI Search, VMs), AWS (Lambda, EKS, ECR, S3, OIDC), Terraform, Helm, Docker, Kubernetes, Caddy, GitHub Actions, Azure DevOps (CI/CD, Self-Hosted Agents), Git, Linux/Bash
- **Professional & Soft Skills:** Technical Mentorship & Teaching, Agile / Scrum Facilitation, Public Speaking & Executive Presentations, Cross-Functional Teamwork, Technical Communication & Data Storytelling, Figma